

Methodological research on partial least squares structural equation modeling (PLS-SEM)

Methodological
research on
PLS-SEM

An analysis based on social network approaches

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Abstract

Purpose – The purpose of this paper is to explore the knowledge infrastructure of methodological research on partial least squares structural equation modeling (PLS-SEM) from a network point of view. The analysis involves the structures of authors, institutions, countries and co-citation networks, and discloses trending developments in the field.

Design/methodology/approach – Based on bibliometric data downloaded from the Web of Science, the authors apply various social network analysis (SNA) and visualization tools to examine the structure of knowledge networks of the PLS-SEM domain. Specifically, the authors investigate the PLS-SEM knowledge network by analyzing 84 methodological studies published in 39 journals by 145 authors from 106 institutions.

Findings – The analysis reveals that specific authors dominate the network, whereas most authors work in isolated groups, loosely connected to the network's focal authors. Besides presenting the results of a country level analysis, the research also identifies journals that play a key role in disseminating knowledge in the network. Finally, a burst detection analysis indicates that method comparisons and extensions, for example, to estimate common factor model data or to leverage PLS-SEM's predictive capabilities, feature prominently in recent research.

Even though this paper does not use the statistical software SmartPLS (www.smartpls.com), Ringle acknowledges a financial interest in SmartPLS.



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Originality/value – Addressing the limitations of prior systematic literature reviews on the PLS-SEM method, this is the first study to apply SNA to reveal the interrelated structures and properties of PLS-SEM's research domain.

Keywords Structural equation modelling, Social network analysis, Partial least squares, PLS-SEM, Co-authorship networks, Knowledge networks

Paper type Research paper

1. Introduction

Since its introduction to applied business research by Wynne W. Chin in the late 1990s (Chin, 1995, 1998), partial least squares structural equation modeling (PLS-SEM) has undergone rapid progress. New developments ranging from new estimators (e.g. Dijkstra and Henseler, 2015; Dolce *et al.*, 2018; Schuberth and Cantaluppi, 2017) and model evaluation metrics (e.g. Aguirre-Urreta and Rönkkö, 2018; Franke and Sarstedt, 2019; Henseler *et al.*, 2015; Sharma *et al.*, 2019) to complementary methods (e.g. Hult *et al.*, 2018; Nitzl *et al.*, 2016; Ringle and Sarstedt, 2016; Schlittgen *et al.*, 2016) and textbooks (Garson, 2016; Hair, Hult, Ringle and Sarstedt, 2017; Hair, Sarstedt, Ringle and Gudergan, 2018; Ramayah *et al.*, 2018) have greatly extended researchers' methodological toolbox and contributed to the emancipation of the PLS-SEM method from covariance-based SEM (e.g. Sarstedt, Ringle, Henseler and Hair, 2014).

Today, PLS-SEM "is no longer an alternative to covariance-based SEM but has transformed into a stand-alone method capable of solving real-world problems" (Latan and Noonan 2017, p. xi). While the method's usage extends to a variety of fields such as agricultural science, engineering, environmental science, medicine and psychology (e.g. Avkiran, 2018a; Willaby *et al.*, 2015), its main area of application is still business research, particularly management information systems (Hair, Hollingsworth, Randolph and Chong, 2017; Shiau and Chau, 2016; Ringle *et al.*, 2012). Some contexts in which researchers in this field have recently used PLS-SEM include analyzing the influencing factors and consequences of social media usage (e.g. Cao *et al.*, 2016; Kim and Min, 2015; Wu *et al.*, 2016), understanding consumer behavior in social commerce (e.g. Lin *et al.*, 2018; Wu and Li, 2018) and evaluating the impact of customer reviews on purchase behavior (e.g. Lee *et al.*, 2011; Lee and Yang, 2015).

In light of PLS-SEM's quick dissemination, numerous studies have reflected on the application of the method using systematic literature reviews (e.g. Ali *et al.*, 2018; Kaufmann and Gaeckler, 2015; Nitzl, 2016; Peng and Lai, 2012; Richter *et al.*, 2016; Usakli and Kucukergin, 2018). While such reviews are helpful to ensure rigorous research and publication practices, they do not allow analyzing the collaboration networks that have shaped the method. More specifically, they do not offer any insights into the structure of scholarly networks that have been formed as a result of the collaborative works of researchers and that shape, generate, distribute and preserve the PLS-SEM domain's intellectual knowledge (Khan and Park, 2013)[1]. Understanding the structures of these networks is important, however, as they set the rules for the network's power game, in which authors, editors and topics joust for authority and influence (Khan and Wood, 2016). Thereby, these network structures influence the content, output and performance of those involved in its boundaries (Vidgen *et al.*, 2007).

To study the structure of knowledge networks, researchers have started using social network analysis (SNA; Wasserman and Faust, 1994). The SNA is a structured way of investigating networks, mapping relationships and exploring knowledge flows among its different entities (Cross *et al.*, 2001). For example, the SNA allows identifying key authors and author groups, which likely have a considerable impact on the community because they enjoy high levels of autonomy in managing their collaboration in the absence of a central force or authority. Trier and Molka-Danielsen (2013, p. 318) note that such authors and author groups are crucial for scientific progress as "their interactions in network structures

provide the main mechanism for knowledge integration.” The SNA furthers understanding of academic collaboration among authors and extends to other network entities such as institutions and countries.

Researchers have used the SNA to reveal knowledge network structures in different contexts, such as information technology management (Khan and Wood, 2016), information technology outsourcing (Swar and Khan, 2013), sharing economy (Fritze *et al.*, 2018), electronic government (Khan and Park, 2013) and social media systems (Khan, 2013). However, prior research has not used the SNA to investigate the PLS-SEM knowledge network. This is surprising considering that research on the PLS-SEM method has a considerable impact on applied research within the management information systems field and way beyond it.

Addressing this gap in research, this study applies the SNA to investigate the PLS-SEM knowledge network based on 84 method-related studies published in 39 journals by 145 authors from 106 institutions, and 25 countries. In doing so, our objective was to shed light on questions such as: Is the PLS-SEM knowledge network fragmented or well-formed? How are certain authors and institutions positioned in the network? Do collaborations span across multiple country boundaries? Has the PLS-SEM method sprawled geographically from its European birth site? Which research themes are trending and which ones are fading? Understanding these questions helps one to identify whether problems and opportunities exist in the way certain authors and institutions collaborate.

To answer these questions, the remainder of the paper proceeds as follows: We first present the data for our analysis and discuss the network types and properties considered in our analysis. Next, we discuss the results of the author, institutions and country networks, followed by the source co-citation networks. In doing so, we highlight the most important entities in each network and point to potential discrepancies when analyzing their network positions from different perspectives (e.g. number of linkages vs their ability to exploit the network). The description of the results from our burst analysis, which allows identifying emerging and fading themes in the PLS-SEM domain, concludes the analysis. Finally, we summarize our main findings, discuss their implications for the field and offer opportunities for further research.

2. Social network analysis

2.1 Data

The data for this research were obtained from the Web of Science (WoS) database[2]. To retrieve the relevant articles, we developed a comprehensive keyword list used in previous PLS-SEM review studies (e.g. Hair, Sarstedt, Pieper and Ringle, 2012; Hair, Sarstedt, Ringle and Mena, 2012; Ringle *et al.*, 2012). We entered the following research query into the WoS search engine to find the publications (from 1965 to early 2017 and across all databases) with the following topics in the title, keywords and abstract[3]:

“Partial least squares structural equation modeling” OR “Partial least squares structural equation modelling” OR “Partial least squares path modeling” OR “Partial least squares path modelling” OR “PLS path modeling” OR “PLS path modelling” OR “path model with latent variables” OR “PLS-SEM” OR “PLS path model” OR “PLS-PM” OR “SmartPLS” OR “PLSgraph” OR “PLS-Graph” OR “XLSTAT” OR “semPLS” OR “matrixpls” OR “ADANCO” OR “PLSgui” OR “PLS-GUT” OR “LVPLS.”

The search retrieved an initial number of 874 articles. Two professors and one post-doctoral researcher proficient in the PLS-SEM method then independently coded all articles to identify methodological research articles. Relevant papers for this study address method comparisons (e.g. Reinartz *et al.*, 2009; Goodhue *et al.*, 2012; Hair, Hult, Ringle, Sarstedt and Thiele, 2017), methodological advances (e.g. Ringle, Sarstedt and Schlittgen, 2014;

Henseler *et al.*, 2012; Henseler, Ringle and Sarstedt, 2016), tutorial articles (e.g. Ingenhoff and Buhmann, 2016; Henseler, Hubona and Ray, 2016a; Nitzl *et al.*, 2016), review articles (e.g. Hair, Sarstedt, Pieper and Ringle, 2012; Hair, Sarstedt, Ringle and Mena, 2012; Kaufmann and Gaeckler, 2015), and theory papers on the method (e.g. Mateos-Aparicio, 2011; Rigdon 2016; Sarstedt *et al.*, 2016). Empirical articles using PLS-SEM without offering a contribution to methods research (e.g. Huma *et al.*, 2017), book reviews (e.g. Ketchen, 2013), and articles dealing with PLS regression (e.g. Krishnan *et al.*, 2011) were excluded from the analysis. The coding agreement on the relevant articles was 93 percent, which compares well with related research (e.g. Shook *et al.*, 2004; Hair, Sarstedt, Pieper and Ringle, 2012; Hair, Sarstedt, Ringle and Mena, 2012; Ringle *et al.*, 2019). Inconsistencies in the coding were resolved via personal discussions among the team of researchers.

This search resulted in 84 articles published in 39 journals. Most of these articles were published in *Journal of Business Research* (ten articles, 11.90 percent), *Long Range Planning* (seven articles, 8.33 percent) and *Industrial Management and Data Systems* (six articles, 7.14 percent). In addition, several statistics journals (e.g. *Computational Statistics*, *Computational Statistics and Data Analysis*, *Journal of Applied Statistics*) have regularly published articles on the PLS-SEM method. This result runs contrary to Rönkkö and Evermann’s (2013, p. 426) notion emphasizing an “absence of articles on PLS in the research methods literature.”

Table I shows the 20 most cited articles considered in the analysis. Interestingly, the top three articles combined have a considerably higher number of citations than those of the remaining 17 articles[4]. Figure 1 illustrates the distribution of citations of all 84 articles over time, showing that the number of citations increased exponentially in the past years.

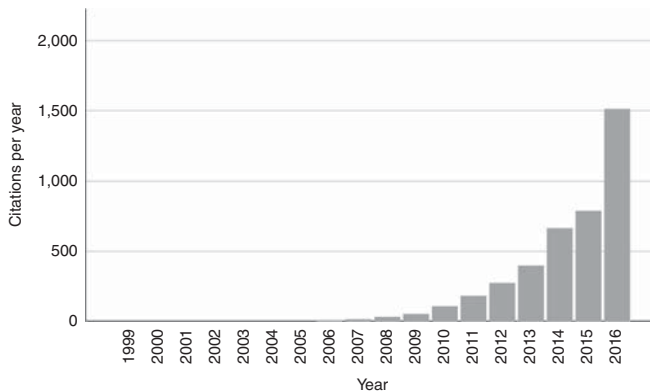
2.2 Network types

We constructed several network types to analyze the knowledge and semantic structures of the PLS-SEM domain based on the previously identified 84 articles. To examine

Rank	Reference	Citations
1	Tenenhaus <i>et al.</i> (2005)	1,367
2	Hair, Sarstedt, Ringle and Mena (2012)	795
3	Wetzels <i>et al.</i> (2009)	755
4	Ringle <i>et al.</i> (2012)	337
5	Hair, Sarstedt, Pieper and Ringle (2012)	170
6	Henseler <i>et al.</i> (2014)	176
7	Henseler and Chin (2010)	157
8	Hair <i>et al.</i> (2013)	160
9	Becker <i>et al.</i> (2012)	155
10	Gudergan <i>et al.</i> (2008)	114
10	Henseler and Sarstedt (2013)	114
12	Henseler, Hubona and Ray (2016b)	82
13	Sarstedt, Ringle, Smith, Reams and Hair (2014)	64
13	Rigdon (2012)	64
15	Sarstedt and Ringle (2010)	63
15	Henseler (2010)	63
15	Dijkstra and Henseler (2015)	63
18	Tenenhaus (2008)	58
19	Rönkkö and Evermann (2013)	55
20	Hsu <i>et al.</i> (2006)	51

Table I.
Top 20 cited methods
papers on PLS-SEM

Note: Citations as of February 5, 2018



Note: $n = 84$

Figure 1.
Citations per year of
all methods papers
on PLS-SEM

connectivity patterns of key publications in the field, we used the SNA to construct author, institution, country and source co-citation networks.

An author network forms when the authors (also referred to as nodes in network terms) publish in journals and establish co-authorship relationships (also referred to as links or ties in network terms). The author network is useful to reveal network structures of scientific collaborations among individual researchers (Liu *et al.*, 2005). To comprehend the author network, we examine and describe the entire network structure on the network level and the specific characteristics on the network's node level (Xu and Chau, 2006; Vidgen *et al.*, 2007; Trier and Molka-Danielsen, 2013, Khan and Wood, 2016). Similarly, an institution network forms based on the affiliations of the authors who publish together. These networks help to understand the establishment of co-authorship ties and the knowledge flow among institutions (Swar and Khan, 2013). Finally, the country network analysis takes these co-authorship ties to a higher level by highlighting the knowledge flow among the institutions' originating countries.

Complementing these three network types, our analysis also considers the source co-citation network, which forms when papers co-cite sources (e.g. journals and conference proceedings) in their reference sections. Researchers use these networks to study relationships and similarities among sources that are foundations of the scientific work (Ding *et al.*, 2000; Tsay *et al.*, 2003). Source co-citation patterns allow disclosing different research groups and schools of thought in the subject area.

We used NodeXL (Smith *et al.*, 2010) and Pajek (Nooy *et al.*, 2005) to analyze and visualize the author and institution networks. To construct the source co-citation network, we used VOSviewer (Van Eck and Waltman, 2010).

2.3 Network properties

A network consists of subnetworks that represent the network components (Hanneman and Riddle, 2005). The core component of a network has the most nodes to other components in the network. While connections exist between the component's nodes, a component may not be connected to other components (Wasserman and Faust, 1994). The connections between the nodes (i.e. the ties) differ in length. The longer a visualized connection (a line), the longer it takes for the information or ideas to pass through the network from one node to another.

A network's diameter is the longest connection or path in a network and as such describes the size of the network (Wasserman and Faust, 1994). The network's density relates to the ratio of established ties to the number of all possible ties in the network.

That is, a density of one describes a fully connected network in which each node connects to every other node. In addition, the clustering coefficient depicts the density of connections in the network in that it indicates the degree of interrelatedness of the network components (Barabási *et al.*, 2002). Finally, the average degree represents the average number of links among the nodes in a network.

In terms of node-level properties, the degree describes the number of a node's ties to other nodes in the network (e.g. an author, institution or country). While the degree of a node is a quantitative description of its relations to other nodes, its betweenness centrality relates to its position in the network. The betweenness centrality can be used to characterize a node's influence or control on collaborations and the flow of information (Liu *et al.*, 2005). Nodes that exhibit a high degree and betweenness centrality are focal points in the network (hubs). They have many connections in the network. The eigenvector centrality quantifies a node's networking ability by considering its connections with other nodes in relation to its importance within these connections (Marsden, 2008).

Established connections in a network facilitate the development of stronger and additional relationships, but also constrain them. As a consequence, the specification of connections in the network determines structural holes. They emerge, for example, when some nodes have an advantageous location in the network that allows them to better form co-authorship ties than other nodes with less expedient locations (Hanneman and Riddle, 2005).

To capture structural holes, we computed each node's aggregate constraint, which is a node's sum of constraints derived from all its ties (Nooy *et al.*, 2005). In comparison with nodes that have low aggregate constraints, nodes with high aggregate constraints have fewer opportunities to form new collaboration ties. They are more constrained to exploit the structural holes in the network (Nooy *et al.*, 2005). Nodes with low aggregate constraints have leeway to leave the network without causing major changes in its overall structure.

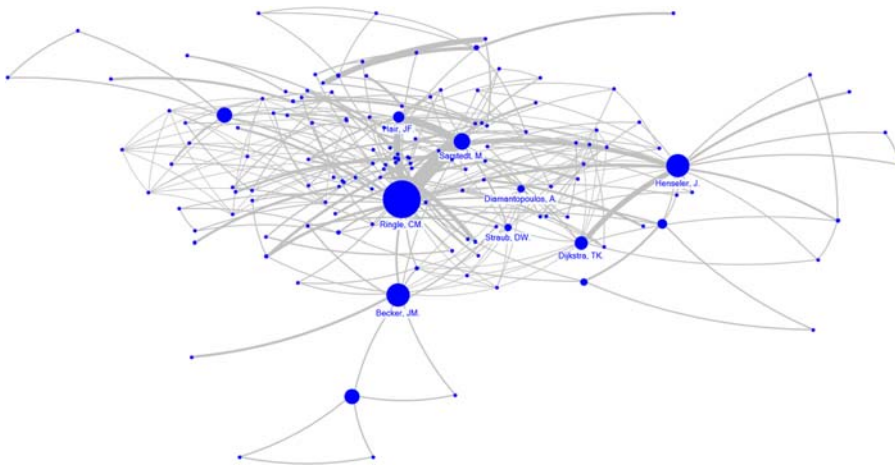
2.4 Burst detection

In a final analysis step, we examine the temporal evolution of emerging topics in the PLS-SEM disciplines. We apply Kleinberg's (2003) burst detection algorithm, which has been extensively used to identify emerging trends in their research domains (e.g. Chen, 2006; Chen *et al.*, 2009; Mane and Borner, 2004). This algorithm employs a probabilistic automaton whose states correspond to the frequencies of individual words. The automaton's state transitions correspond to points in time around which the frequency of the word changes significantly. Using the articles' titles and abstracts as input, the burst detection identifies words or word pairs that experience a sudden increase in usage frequency along with the beginning and ending of the burst. The burst weight indicates the change in usage frequency (Guo *et al.*, 2011). To run the burst detection analysis, we use Sci2Team's (2009) Science of Science tool.

3. Author network

3.1 Network-level analysis

Overall, 145 authors participated in the network to form 278 co-authorship ties. Figure 2 visualizes the network structure where nodes represent the authors, the links among the nodes indicate the co-authorship relationships, and the node sizes represent each author's betweenness centrality. The network's average degree (the average number of co-authors a person has published with) is 3.83. The network comprises 35 connected components (subnetworks) with two or more authors and seven isolates (solo authors). The largest connected component comprises 48 authors (33.10 percent of all authors), whereas the second largest component only comprises nine authors (6.21 percent authors), followed by the third and fourth largest components with seven authors (4.83 percent authors) and five authors (3.45 percent authors). The Appendix reports the members of the four largest components in the author network.



Notes: Node sizes indicate betweenness centrality; link widths indicate collaboration intensity; only authors with ten or more ties are shown

Source: Created with NodeXL Pro (<http://nodexl.codeplex.com>) from the Social Media Research Foundation (www.smrfoundation.org)

Figure 2.
Author collaboration
network

In the central component, the network appears to be dominated by a few researchers who have established densely connected sections around them. The rest of the network is rather fragmented, with several isolated clusters of authors working in silos, as evidenced in the high number of components relative to the number of co-authors and the network's density, which indicates that only 3 percent of all possible network ties have been realized. While this density is higher than in related research (e.g. Khan and Park, 2013; Khan and Wood, 2016; Swar and Khan, 2013), it is still low in absolute terms.

Similarly, the clustering coefficient of 0.70 is much higher than the network's density, indicating that the authors are embedded in dense clusters, with limited ties outside local neighborhoods (clusters). The network's small diameter (2.1) also suggests that the authors contributing in the domain have a high tendency to form groups. Overall, this structure is very similar to authorship networks encountered in the information systems field (Trier and Molka-Danielsen, 2013; Vidgen *et al.*, 2007; Xu and Chau, 2006).

3.2 Node-level analysis

Table II shows the top 20 authors in terms of the degree, betweenness centrality, eigenvector centrality and aggregate constraints. The results in terms of degree and betweenness identify Jörg Henseler, Christian M. Ringle and Marko Sarstedt as hubs in the network, characterized by many connections with other nodes in the network. In addition, Jan-Michael Becker, Theo K. Dijkstra and Joseph F. Hair also serve as focal points in the network. When analyzing the authors in terms of the importance of their connections (the eigenvector centrality), we find that Theo K. Dijkstra, Christian M. Ringle and Marko Sarstedt reside in prominent network positions. Contrasting these results with the network structure in Figure 2 shows that some authors who score high on betweenness centrality have comparatively few ties with other nodes in the network (e.g. Gabriel Cepeda Carrión, Siegfried P. Gudergan, José Roldán and Martin Wetzels). In other words, these authors have good information control over the network, despite their smaller number of ties compared to other authors.

Table II.
Top 20 authors

Rank	Degree	Betweenness centrality	Eigenvector centrality	Aggregate constraint
1	Ringle, C.M.	Ringle, C.M.	Ringle, C.M.	Becker, J.-M.
2	Sarstedt, M.	Becker, J.-M.	Sarstedt, M.	Henseler, J.
3	Henseler, J.	Henseler, J.	Dijkstra, T.K.	Ringle, C.M.
4	Hair, Joseph F.	Sarstedt, M.	Straub, D.W.	Diamantopoulos, A.
5	Dijkstra, T.K.	Gudergan, S.P.	Diamantopoulos, A.	Rigdon, E.E.
6	Straub, D.W.	Wetzels, M.	Hair, J.F.	Karahanna, E.
7	Diamantopoulos, A.	Dijkstra, T.K.	Henseler, J.	Rai, A.
8	Becker, J.-M.	Hair, J.F.	Ketchen, D.J.	Hult, G.T.M.
9	Ketchen, D.J.	Roldán, J.L.	Hult, G.T.M.	Calantone, R.J.
10	Hult, G.T.M.	Straub, D.W.	Calantone, R.J.	Ketchen, D.J.
11	Calantone, R.J.	Diamantopoulos, A.	Becker, J.-M.	Gudergan, S.P.
12	Zhao, J.	Cepeda Carrión, G.	Rai, A.	Straub, D.W.
13	Zhu, D.	Esposito Vinzi, V.	Rigdon, E.E.	Roldán, J.L.
14	Zhang, X.	Richter, N.F.	Karahanna, E.	Dijkstra, T.K.
15	Yuan, Z.	Fassott, G.	Gudergan, S.P.	Li, F.
16	Yang, X.	Rai, A.	Thiele, K.O.	Yang, X.
17	Xue, F.	Tenenhaus, M.	Smith, D.	Yuan, Z.
18	Peng, B.	Schlittgen, R.	Reams, R.	Zhang, X.
19	Liu, Y.	Evermann, J.	Pieper, T.M.	Liu, Y.
20	Li, F.	Ketchen, D.J.	Mena, J.A.	Peng, B.

Figure 3 shows the distribution of the nodes' aggregate constraint values, indicating that many authors have an aggregate constraint of around 1.0 (mean = 0.813; standard deviation = 0.257). In terms of the structural holes, the analysis revealed 12 authors (8.3 percent of all 145 authors) with very low aggregate constraints, ranging from 0.223 to 0.385. Specifically, Jan-Michael Becker has the lowest aggregate constraint, followed by Jörg Henseler and Christian M. Ringle (Table II). These authors, therefore, have the greatest opportunities to exploit the structural holes due to their position in the network. On the contrary, 16 authors (12.41 percent of all authors) have fewer opportunities to form new collaborations, as evidenced in their high aggregate constraints values above 1.0. Maurizio Carpita has the highest aggregate

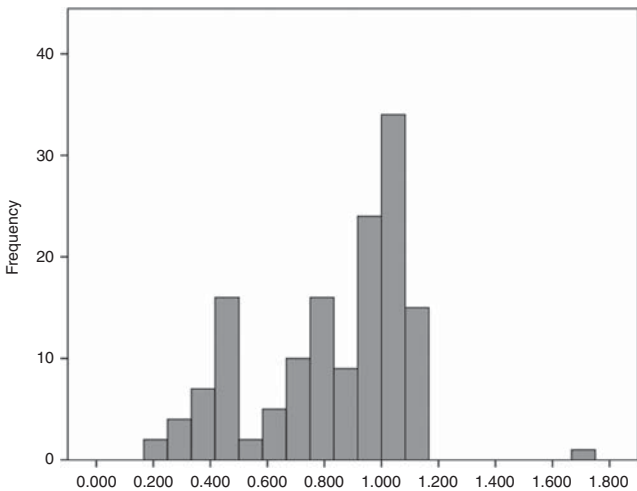


Figure 3.
Aggregate constraint
values

Note: n = 145

constraint value. Other authors in this tier include John Antonakis, Jeffrey R. Edwards and Cameron M. McIntosh, who critically commented on the PLS-SEM method (McIntosh *et al.*, 2014). While these authors may not take a prominent position in the network, their contributions to the literature on PLS-SEM are important, as they point to problems in other researchers' understanding and use of the method, triggering substantial follow-up research (e.g. Henseler *et al.*, 2014, 2015; Henseler, Ringle and Sarstedt, 2016).

Overall, the results suggest that only a few authors have a position that allows them to utilize the network, while most authors did not considerably benefit from their network position. More specifically, these authors could not form collaboration ties with other authors located in isolated clusters. Therefore they could not use their existing network ties to obtain information and control advantages over other authors (Burt, 1992).

This result does not suggest that these authors could not form collaboration ties at all. Even though their position in the network makes it more difficult to form ties based on previous co-authorships, they could form collaboration ties through means other than previous co-authorships, such as conferences relevant to the discipline (e.g. The International Conference on PLS and Related Methods). Finding such a network structure is not entirely surprising, considering that methodological developments of PLS-SEM are a relatively new field of research (Latan and Noonan, 2017).

4. Institutions network

4.1 Network-level analysis

Overall, 106 institutions participated in the network to form 226 co-authorship ties. The network has 31 connected components (with at least two nodes) and 11 isolates (publishing institutions that have no co-authorship ties with other institutions). The largest connected component comprises 44 institutions (41.51 percent of all institutions), forming 175 ties. The average degree (the average number of institutions an institution has published with) is 3.85, the density is 4 percent, the diameter is 4, and the average clustering coefficient is 0.56. The institution network, therefore, has a similar structure as the author network, but with a broader base of institutions in the central component.

Figure 4 illustrates the resulting institution network. The nodes represent institutions with node sizes indicating each node's betweenness centrality. The links connecting the nodes represent co-authorship ties, whereby thicker links indicate a stronger collaboration. The network graph shows particularly strong ties between Hamburg University of Technology, Kennesaw State University, the University of Magdeburg, and the University of Newcastle in Australia. This is not surprising, considering the strong co-authorship ties among these institutions' researchers, such as Joseph F. Hair, Christian M. Ringle and Marko Sarstedt, who also rank highly in the author network (Table II).

4.2 Node-level analysis

Next, we focus on the top 20 institutions in terms of degree, betweenness centrality, eigenvector centrality and aggregate constraints (Table III). The results closely mirror those of the author network. The affiliations of authors, who serve as hubs in the network, generally exhibit high levels of degree, betweenness centrality and eigenvector centrality. The Hamburg University of Technology and the University of Newcastle stand out in terms of their importance for the network, followed by the University of Magdeburg and the Kennesaw State University. Moreover, the former two institutions exhibit low aggregate constraints, indicating that their members are in a good position to exploit structural holes in the network. The results also show that Georgia State University has fewer node connections in the network, but the existing connections are important for the network structure.

When important authors change affiliations, it influences the institution network. For example, in 2014 Jörg Henseler moved from the Radboud University Nijmegen to the University

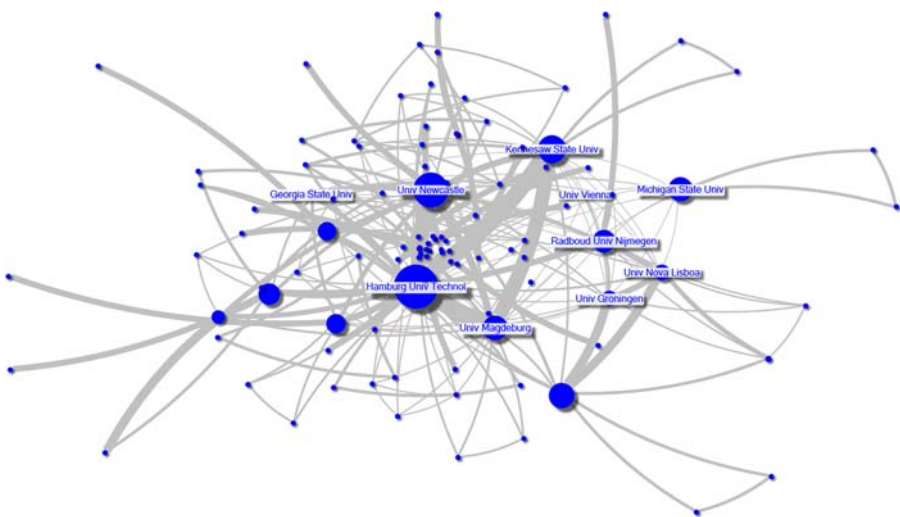


Figure 4.
Institutional
collaboration network

Notes: Node sizes indicate betweenness centrality; link widths indicate collaboration intensity; only the top ten institutions in terms of degree of are labeled (Table III)

Source: Created with NodeXL Pro (<http://nodexl.codeplex.com>) from the Social Media Research Foundation (www.smrfoundation.org)

of Twente. Whereas his earlier publications at the Radboud University Nijmegen entailed a low aggregate constraint value for this institution, his recent works moved the University of Twente into a position to influence collaborations and flow of information (eigenvector centrality). Similar findings hold for Marko Sarstedt, who moved from the Ludwig-Maximilians-University Munich to the University of Magdeburg in 2012, and for Joseph F. Hair, who moved from Kennesaw State University to the University of South Alabama in 2016.

Finally, we expect a significant drop of the University of Newcastle's relevance in the future since Siegfried P. Gudergan moved from this institution to the University of Waikato in New Zealand, while Christian M. Ringle and Marko Sarstedt moved their adjunct professor roles from the University of Newcastle to the University of Waikato and Monash University of Malaysia, respectively.

Further analysis reveals that 19 institutions (17.92 percent of all institutions) have low aggregate constraint values between 0.18 and 0.39, and are therefore in a good position to exploit structural holes in the network. Whereas 23 institutions (21.70 percent of all institutions) have medium aggregate constraint values of up to 0.71, most of them (64 institutions; 60.38 percent of all institutions) have very high constraint values up to 1.12. Overall, the network results suggest that only a few institutions (19 institutions; 17.92 percent) were positioned well to exploit the network and that the majority (64 institutions; 60.38 percent) has difficulty using their position to benefit from the network.

5. Country network

5.1 Network-level analysis

Figure 5 shows the country-level collaboration for coauthoring and publishing methodological research on PLS-SEM. The network comprises 25 countries and 86 co-authorship ties. The network's density is 14.33 percent, indicating that only few of all possible links have been established in the network. For example, Belgium, Indonesia and Taiwan appear as isolates in the network. Similarly, France, Italy and Sweden form a subnetwork, which is not linked to

Rank	Degree	Betweenness centrality	Eigenvector centrality	Aggregate constraint
1	Hamburg University of Technology	Hamburg University of Technology	Hamburg University of Technology	University of Newcastle
2	University of Newcastle	University of Newcastle	University of Newcastle	Radboud University Nijmegen
3	University of Magdeburg	Kennesaw State University	University of Magdeburg	Hamburg University of Technology
4	Kennesaw State University	University of Twente	Kennesaw State University	University of Magdeburg
5	Nova University of Lisbon	University of Magdeburg	Georgia State University	Auburn University
6	University of Groningen	Michigan State University	University of Groningen	Ludwig-Maximilians-University Munich
7	Radboud University Nijmegen	Radboud University Nijmegen	Nova University of Lisbon	University of Vienna
8	Georgia State University	Ludwig-Maximilians-University Munich	Radboud University Nijmegen	University of Twente
9	Michigan State University	University of Cologne	University of Vienna	University of Cologne
10	University of Vienna	University of Seville	Michigan State University	Nova University of Lisbon
11	University of Twente	Nova University of Lisbon	Auburn University	University of Groningen
12	University of Cologne	University of Groningen	University of Twente	Georgia State University
13	Auburn University	University of Hamburg	University of Cologne	University of Seville
14	Ludwig-Maximilians-University Munich	Georgia State University	Ludwig-Maximilians-University Munich	University of Hamburg
15	University of Hamburg	University of Vienna	Oglethorpe University	Michigan State University
16	University of Seville	University of Kaiserslautern	Ryerson University	Kennesaw State University
17	University of Kaiserslautern	Memorial University of Newfoundland	University of Hamburg	Villanova University
18	Oglethorpe University	Auburn University	Middle Tennessee State University	University of Manchester
19	Ryerson University	Oglethorpe University	University of Manchester	Ryerson University
20	University of Manchester	University of Manchester	Nordakademie	Nordakademie

Table III.
Top 20 institutions

the main network (Figure 5). On the contrary Australia, Germany and the USA appear as focal countries in the network. The average degree (the average number of institutions that an institution has published with) is 3.440 and the diameter is 2. Finally, the network's average clustering coefficient is 0.336, which is higher than the density of the network, indicating the cliquishness of the network (Swar and Khan, 2013).

5.2 Node-level analysis

Table IV shows the top 20 countries in terms of degree, betweenness centrality, eigenvector centrality and aggregate constraints. The top 3 countries with the highest degree and eigenvector centrality are Germany, Australia and the USA, mirroring their importance in the network. Similarly, Germany, the USA and Australia are the top 3 countries in terms of betweenness centrality. Finally, the country ranking based on the aggregate constraints shows a slightly different picture with Canada and Spain having the smallest aggregate constraint, followed by Germany, Australia and Portugal.

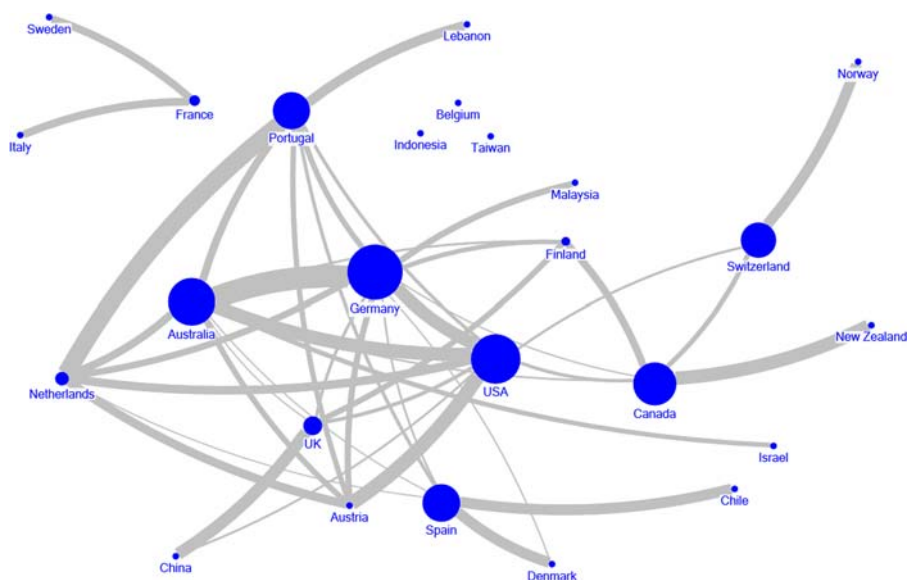


Figure 5.
Country collaboration
network

Source: Created with NodeXL Pro (<http://nodexl.codeplex.com>) from the Social Media Research Foundation (www.smrfoundation.org)

Rank	Degree	Betweenness centrality	Eigenvector centrality	Aggregate constraint
1	Germany	Germany	Germany	Canada
2	Australia	USA	Australia	Spain
3	USA	Australia	USA	Germany
4	Portugal	Canada	Portugal	Australia
5	Spain	Spain	The Netherlands	Portugal
6	Canada	Portugal	Austria	USA
7	The Netherlands	Switzerland	Spain	Finland
8	UK	UK	Canada	UK
9	Austria	The Netherlands	UK	Switzerland
10	Finland	France	Finland	The Netherlands
11	Switzerland	Finland	Denmark	France
12	China	Austria	Switzerland	Austria
13	France	China	China	Denmark
14	Denmark	Denmark	Malaysia	China
15	Norway	Norway	Israel	New Zealand
16	Lebanon	Lebanon	Lebanon	Norway
17	Malaysia	Malaysia	Chile	Indonesia
18	Sweden	Sweden	New Zealand	Taiwan
19	Italy	Italy	Norway	Israel
20	Chile	Chile	France	Sweden

Table IV.
Top 20 countries

6. Source co-citation networks

In the next step, we analyzed the relationships and similarities among journals publishing methodological PLS-SEM research by examining their source co-citation networks. These networks form when papers co-cite sources (e.g. journals and conference proceedings)

in their reference lists. Out of the total sources cited ($n=1,103$) of the 84 articles, we considered only sources that were cited at least ten times ($n=95$) in the analysis.

Table V shows the top 10 journals in terms of network properties and co-citations[5]. The network-level properties indicate that, in terms of degree, betweenness centrality, and eigenvector centrality, *MIS Quarterly*, *Long Range Planning*, the *Journal of Marketing Research*, the *Journal of the Academy of Marketing Science* and *Advances in International Marketing* were the most influential outlets and the central key players in terms of quality of information flow in the network. For example, in 2013 and 2014, *Long Range Planning* published a series of special issues on PLS-SEM (Hair, Ringle and Sarstedt, 2012; Hair *et al.*, 2013, 2014), which contained review articles (Hair, Sarstedt, Pieper and Ringle, 2012), commentaries (Bentler and Huang, 2014; Dijkstra, 2014; Rigdon, 2012, 2014; Sarstedt, Ringle, Henseler and Hair, 2014), and methods articles (e.g. Becker *et al.*, 2012). Similarly, *Long Range Planning* and *MIS Quarterly* featured several highly cited articles on the method (Table I).

In a further analysis, we evaluated the frequency with which two sources were co-cited. The results in Table V indicate that the *Journal of Marketing Research*, *Long Range Planning* and *MIS Quarterly* are frequently co-cited. Similarly, *MIS Quarterly*, *Multivariate Behavioral Research* and *Psychometrika* frequently appear jointly in the analyzed articles' reference lists.

The heat map in Figure 6 visualizes the source co-citation network. The intensity of the color reflects whether or not sources are co-cited together, showing that *MIS Quarterly* and the *Journal of Marketing Research* are frequently co-cited. The heat map also shows several isolates, such as the *Journal of Operations Management*, which published a review study (Peng and Lai, 2012) and a particularly critical comment on the method (Rönkkö *et al.*, 2016). Similarly, several specialist journals, such as *Tourism Management* and the *Journal of Purchasing and Supply Management*, appear as isolates. With the increasing dissemination of PLS-SEM into other fields and methodological streams, we expect the core co-citation network to include journals such as *Decision Science* and the *Journal of Retailing*.

7. Burst detection

In the first step of the burst detection analysis, we extracted the top 30 latest bursting and disappearing topics using Kleinberg's (2003) algorithm. As this analysis produced several generic words such as "aim," "identification," and "research," two professors proficient in PLS-SEM then reviewed the list in terms of relevance. As a result, 20 words were eliminated from the list.

Rank	Degree	Betweenness centrality	Eigenvector centrality	Co-citation	
1	MISQ	MISQ	MISQ	JMR	MISQ
2	LRP	LRP	LRP	LRP	MISQ
3	JMR	JMR	JMR	MBR	PM
4	JAMS	JAMS	JAMS	ISR	MISQ
5	AIM	AIM	AIM	MISQ	PM
6	SMJ	HPLS	JMTP	MISQ	SEM
7	PM	PM	SMJ	JBR	MISQ
8	JMTP	ESS	PM	JBR	LRP
9	HPLS	SMJ	ESS	JAMS	LRP
10	ESS	JMTP	HPLS	HPLS	MISQ

Notes: AIM, *Advances in International Marketing*; ESS, *Encyclopedia of Statistical Sciences*; HPLS, *Handbook of Partial Least Squares*; ISR, *Information Systems Research*; JAMS, *Journal of the Academy of Marketing Science*; JBR, *Journal of Business Research*; JMR, *Journal of Marketing Research*; JMTP, *Journal of Marketing Theory & Practice*; LRP, *Long Range Planning*; MBR, *Multivariate Behavioral Research*; MISQ, *MIS Quarterly*; PM, *Psychometrika*; SEM, *Structural Equation Modeling*; SMJ, *Strategic Management Journal*

Table V.
Top ten sources'
co-citation based
on strength

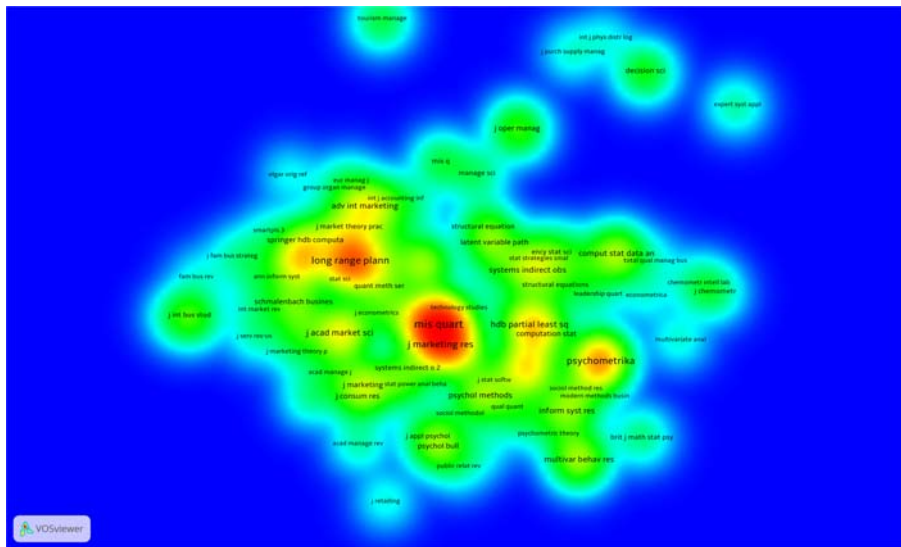


Figure 6.
Source co-citation
network heat map

Note: Only sources cited at least ten times considered ($n=95$)

The burst detection analysis underlines the ongoing relevance of research comparing PLS-SEM's parameter accuracy with those of other SEM methods through the terms “comparison” (weight = 3.22) and “parameters” (weight = 2.51). These terms first appeared in 2015 with a pronounced weight and are still active.

Relatedly, the burst detection analysis underlines the growing interest in the nature of measurement issues in PLS-SEM. For example, Dijkstra and Henseler (2015) and Kock (2017) proposed methods to conduct factor-based PLS-SEM analyses, which adjust the parameter estimates from the standard PLS-SEM algorithm to estimate common factor model data. A different stream of research questions the universal validity of the factor model as assumed

Table VI.
The top 10 latest
bursting and
disappearing topics
in article titles
and abstracts

Rank	Word	Weight	Start	End
1	Comparison	3.22	2015	
2	Identification	2.64	2016	
3	Parameters	2.51	2015	
4	Prediction	2.32	2016	
5	Tutorial	1.88	2016	
6	Consistent	1.19	2015	
7	Small	1.19	2015	
8	Composites	1.10	2016	
9	Complexity	0.89	2014	
10	Fit	0.62	2013	2013

by critics of the PLS-SEM method (e.g. Rönkkö and Evermann, 2013; Rönkkö *et al.*, 2016), instead emphasizing PLS-SEM's nature as a composite-based approach to SEM (e.g. Rigdon, 2012; Rigdon *et al.*, 2017). These streams are represented in the burst analysis through the terms “consistency” (weight = 1.19) and “composites” (weight = 1.10), both of which are still active.

The term “fit” burst in 2013, potentially triggered by Henseler and Sarstedt's (2013) article on Tenenhaus *et al.*'s (2005) goodness-of-fit index and, somewhat surprisingly, disappeared the same year with a rather low weight of 0.62. In contrast to this result, recent research has again raised the concept of fit in a PLS-SEM context. For example, Henseler, Hubona and Ray (2016a, b), Henseler, Ringle and Sarstedt (2016) and Henseler (2017) discuss several measures of approximate model fit that quantify the discrepancy between the model-implied and the empirical correlation matrix. Other researchers also discuss model fit in a PLS-SEM context, noting that because of its prediction focus, fit is less relevant to PLS-SEM compared to covariance-based SEM (e.g. Sarstedt *et al.*, 2017). Instead, researchers should focus on assessing a PLS path model's out-of-sample predictive accuracy (e.g. Evermann and Tate, 2016; Shmueli *et al.*, 2016). These streams of research are identified in the burst analysis through the term “prediction”, whose weight of 2.32 indicates a pronounced frequency of this topic.

The burst detection analysis also highlights the relevance of the topics of small sample sizes (weight = 1.19) and model complexity (weight = 0.89), which many researchers used to justify their choice of the PLS-SEM method (e.g. Ali *et al.*, 2018; Hair, Sarstedt, Pieper and Ringle, 2012; Hair, Sarstedt, Ringle and Mena, 2012). Tutorials (weight = 1.88) that discuss the assessment of PLS-SEM results – also considering the different perspectives on the method – still feature prominently in methods research.

8. Discussion

8.1 Summary of main findings

Initiated by Herman O. A. Wold in the late 1960s (e.g. Wold, 1964, 1965, 1966), PLS-SEM has developed extensively over the last decade (e.g. Avkiran, 2018b; Garson, 2016; Ramayah *et al.*, 2018). Not only has the usage of the method disseminated into a broad range of non-business disciplines such as agricultural science, environmental science, medicine and psychology, but methodological research on PLS-SEM has brought the method to a new level by advancing our understanding of its properties and limitations. Advanced analysis techniques developed over the last decade offer researchers more freedom in their modeling efforts and allow a more nuanced assessment of results (e.g. Franke and Sarstedt, 2019; Hair *et al.*, 2018; Henseler, Ringle and Sarstedt, 2016). All these developments were accompanied by controversies, sometimes even questioning the method's *raison d'être* (e.g. Rönkkö *et al.*, 2016). The debates also led to a diversification of the PLS-SEM community, with differing viewpoints on the nature of measurement, the role of model fit and the method's scope of application (Henseler, 2018).

With these developments in mind, this paper explores the knowledge infrastructure of methodological research on PLS-SEM. Employing the SNA, we investigate the domain's knowledge infrastructure of author, institution and country networks. We find that a compact group of authors dominates the network, whereas most authors work in isolated groups, loosely connected to the network's focal authors. Our analysis of structural holes supports this result, and we show that only a few authors are in the position to utilize the network. Most authors could not form collaboration ties with other authors located in other network clusters, and could therefore not use their existing network connections to obtain information and control advantages over other authors (Khan and Wood, 2016). The institution network analysis mirrors these results on the network as well as the node level. Compared to the author network, the institution network appears to be slightly better formed, as evidenced in its greater diameter and lower clustering coefficient.

Overall, these network structures correspond to those found in other disciplines (Trier and Molka-Danielsen, 2013; Vidgen *et al.*, 2007; Xu and Chau, 2006) and are expected in light of the field's recency. Finally, the country network analysis documents the central role that Australia, Germany and the USA play for the network, indicating that the PLS-SEM method has successfully sprawled geographically from its European birth site.

The source co-citation network analysis indicates that *MIS Quarterly*, *Long Range Planning*, *Journal of the Academy of Marketing Science*, *Journal of Marketing Research* and *Advances in International Marketing* play a key role in disseminating knowledge in the network. Several of these journals are frequently co-cited, indicating that they are intellectually similar in nature (Small, 1973) and that their articles share a similar theme that is relevant to the network.

The results of the burst detection analysis highlight potential areas of interest that are in the focus of today's contemporary research. Specifically, we find that method comparisons and extensions, for example, to estimate common factor model data or to leverage PLS-SEM's predictive capabilities, feature prominently in recent research. The high frequencies with which these concepts emerge, suggest that research interest in these areas will likely continue. On the contrary, the use of small sample sizes in PLS-SEM, which prior research has discussed extensively (e.g. Goodhue *et al.*, 2012; Marcoulides *et al.*, 2012), and the role of model complexity are less relevant in recent methodological research.

8.2 Limitations and future research

While the results of our analysis offer unique insights into the knowledge networks of the PLS-SEM domain, it is important to stress the limited spread of the network, due to the field's recency. As Latan and Noonan (2017, p. xi) note, "the period from that time until the late 1980s can be seen as the "gestation period," which was followed by continued development and especially rapid development in the past decade." Correspondingly, our SNA draws on a relatively small number of nodes compared to other studies. Specifically, the author network in our study considers only 145 authors. This is considerably smaller than, for example, Khan and Wood (2016) with 1,914 authors, Trier and Molka-Danielsen (2013) with 1,360 authors and Xu and Chau (2006) with 1,862 authors. Furthermore, our analysis is limited to English language publications and, therefore, disregards methodological contributions published in other languages (e.g. Sarstedt and Ringle, 2008; Scholderer and Balderjahn, 2006; Ringle, da Silva and de Souza Bido, 2014). Nevertheless, our study can act as a baseline study that subsequent analyses can build upon to trace the field's development over time.

Future research should also consider using SNA more routinely to analyze and synthesize emerging or mature topics in the PLS-SEM domain and related fields, such as composite-based SEM in general. As Khan and Wood (2016, p. 388) note, "when used to synthesize the existing literature from a network perspective, the SNA technique can reveal valuable invisible patterns that can certainly facilitate theory development and uncover areas for future research." We echo this observation in that our analysis goes well beyond the scope of systematic literature review methods, as it can reveal hidden structures in the domain. While this research analyzes the dissemination of PLS-SEM by addressing structures of author, institution and source co-citation networks, and identifies trending schemes in the field, future research may disclose and compare the discipline-specific diffusion of PLS-SEM.

Finally, we hope that our study facilitates building bridges among seemingly disjoint perspectives on the PLS-SEM method by emphasizing the unifying elements of our work. The fact that even the fiercest critics of the methods such as Aguirre-Urreta and Rönkkö (2018) have recently started publishing research designed to advance the PLS-SEM method gives rise to optimism toward a more constructive exchange.

Notes

1. Following prior research (e.g. Khan, 2013; Khan and Wood, 2016; Trier and Molka-Danielsen, 2013), we refer to these networks as knowledge networks in the sense that players form collaboration ties to create new knowledge in the form of publications.
2. The WoS database covers more than 33,000 journals whose inclusion depends on the journals' publishing standards, editorial content, international diversity of authorship and editorial board, and citation data.
3. This search was carried out on April 16, 2017.
4. Note that this analysis only considers articles whose journals were listed in the WoS at the time of publication. For example, Hair *et al.* (2011) published in *Journal of Marketing Theory and Practice* is with far more than 5,000 citations (Google Scholar) one of the most highly cited articles on the PLS-SEM method. It does not appear in the list because the journal was only listed in the WoS in 2015. The source-co-citation analysis in Section 5, however, considers all journals listed in the WoS at the time of the analysis (i.e. also *Journal of Marketing Theory and Practice*).
5. We chose this cut-off because we observe a significant drop in degree, betweenness centrality and eigenvector centrality for lower ranked journals.

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Appendix. Component members in author network (top four components)

First component

Becker, J.-M.; Bouncken, R.; Calantone, R.J.; Cepeda-Carrión, G.; Chin, W.W.; Coelho, P.S.; Diamantopoulos, A.; Dijkstra, T.K.; Fassott, G.; Gudergan, S.P.; Hair, J.F.; Henseler, J.; Hubona, G.; Hult, G.T.M.; Ismail, I.R.; Karahanna, E.; Ketchen, D.J.; Klein, K.; Lorscheid, I.; Matthews, L.M.; Mena, J.A.; Meyer, M.; Nitzl, C.; Odekerken-Schröder, G.; Pieper, T.M.; Rai, A.; Ratzmann, M.; Ray, P.A.; Reams, R.; Richter, N.F.; Rigdon, E.E.; Ringle, C.M.; Roldán, J.L.; Sarstedt, M.; Schlägel, C.; Schlittgen, R.; Schubring, S.; Sinkovics, R.R.; Smith, D.; Straub, D.W.; Taylor, C.R.; Thiele, K.O.; van Oppen, C.; Völckner, F.; Wende, S.; Wetzels, M.; Will, A.; Wilson, B.

Second component

Li, F.Y.; Liu, Y.X.; Peng, B.; Xue, F.Z.; Yang, X.W.; Yuan, Z.S.; Zhang, X.S.; Zhao, J.H.; Zhu, D.W.

Third component

Chatelin, Y.M.; Davino, C.; Esposito Vinzi, V.; Lauro, C.; Squillacciotti, S.; Tenenhaus, M.; Trinchera, L.

Fourth component

Chen, X.H.; Chen, Y.; Liu, B.S.; Shen, Y.H.; Sun, H.

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